CHANDIGARH UNIVERSITY
Discover. Learn. Empower.

Experiment 03

Name	: Johnson Kumar	**UID**	: 23BCS12654
Branch	: CSE	**Section**	: KRG-1A
Semester	: Sixth	**Date of exp.**	: 27/01/26
Subject	: System Design	**Subject Code**	: 23CSH-314

AIM

To design a social media platform similar to Facebook or Instagram.

OBJECTIVE:

- To enable secure user registration and login for account management.
- To allow content creation and sharing such as photos, videos, and stories.
- To provide a personalized feed based on user activity and connections.
- To support social interactions including likes, comments, follows, and messaging.
- To ensure scalability and reliability for handling large numbers of users.
- To maintain data security and privacy while protecting user information.

REQUIREMENTS

Functional Requirements:

- Client should be able to register and login to the application.
- Client should be able to create post (text / image / videos).
- Client should be able to follow each other (or send friend requests).
- Client should be able to like or comment on the post.
- Client should be able to view the feed of post from users they follow.

Non-Functional requirements:

- Scalability: 500M DAU
- Consistency & Availability:
  We need our application to be HIGHLY available, then consistent.
  Reason: If it is not operational / Functional when it was required, then there is no meaning of developing the application.

- Latency: (Uploading speed of publishing post): 500ms to upload post

HIGH LEVEL DESIGN

Core-Entities

- Users
- Post
- Followers
- Like & Comment
- Feeds

Attached socialmedia.drawio file has whole high-level design.

LOW LEVEL DESIGN

API-Endpoints

- User On-boarding API's:
  a. User Registeration: POST API CALL : POST / api / users / register_user
  b. User Login: POST API CALL: POST / api / users / login
  c. User Data Display: GET API CALL: GET / api / users / {user_id} / profile
  d. User Data Update: PUT API CALL: PUT / api / users / {user_id} / profile

- User Post's:
  a. POST / api / user_id / posts
  b. GET / api / posts / {post_id}
  c. PUT / api / posts / {post_id}
  d. DELETE / api / posts / {post_id}
  e. GET / api / posts / feed / limit = {limit} & offset = {offset}: PAGINATION
  f. GET / api / users / {user_id} / posts: PAGINATION

- C. User Interactions
  a. POST / api / posts / {post_id} / like
  b. DELETE / api / posts / {post_id} / unlike
  c. POST / api / posts / {post_id} / comments
  d. GET / api / posts / {post_id} / comments
  e. PUT / api / comments / {comment_id}
  f. DELETE / api / comments /post_idy {comment_id}
  g. POST / api / users / {user_id} / follow
  h. DELETE / api / users / {user_id} / unfollow

LEARNING OUTCOMES

- I learned how to design a social media platform using multiple databases like PostgreSQL for relational data, Cassandra for distributed NoSQL storage, and S3 for media storage.
- I got to know the importance of caching tools like Redis to improve performance and reduce latency in feed generation.
- I learned how Kafka can be used for event streaming and handling real-time notifications or activity logs.
- I understood how to integrate different storage and database systems to balance scalability, reliability, and speed.
- I got to know how microservices interact with these tools to deliver features like user management, content sharing, and personalized feeds.
- I learned the role of infrastructure components (API gateway, load balancer, CDN) in ensuring smooth delivery of content to millions of users.